

Content Publishing + Full-Text Search Using PostgreSQL

DSCI 551 - Foundations of Data Management

Phase 1 Project Proposal

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Database Selection and Motivation:

For this project, I will use PostgreSQL as the database system. PostgreSQL is a widely used open-source relational database known for its extensibility and advanced indexing mechanisms. It provides built-in full-text search functionality and specialized index types, such as Generalized Inverted Indexes (GIN), making it suitable for efficient keyword-based search applications.

PostgreSQL is chosen because its internal text-search architecture and GIN indexing structure allow meaningful analysis of how internal storage and indexing decisions affect application-level search performance.

Internal Architecture Focus:

As this is an individual project, the primary internal focus will be on PostgreSQL's full-text search internals and Generalized Inverted Index (GIN) structure.

This includes analyzing:

- How PostgreSQL converts text into `tsvector` representations
- How GIN indexes store and retrieve tokenized terms
- How the query planner processes full-text search queries using the `@@` operator
- Performance differences between indexed search and sequential text scans

This focus is selected because GIN indexing differs significantly from traditional B-tree indexing and provides an opportunity to study inverted index behavior within a relational database system, particularly in search-oriented workloads.

Proposed Application:

The application will be a lightweight article search platform. Users will be able to:

- Create and update articles

- Search articles using keyword-based queries
- Retrieve ranked search results

A representative query for analysis is shown below:

```
SELECT id, title
FROM articles
WHERE to_tsvector('english', body)
      @@ to_tsquery('database & system')
ORDER BY published_at DESC
LIMIT 20;
```

This application will demonstrate how PostgreSQL's full-text search engine and GIN indexing improve search efficiency compared to sequential scans.

Responsibilities:

As this is an individual project, I will be responsible for:

- Installing and configuring PostgreSQL
- Designing the database schema and data model
- Implementing tsvector generation for text columns
- Creating and evaluating GIN indexes
- Conducting performance experiments using EXPLAIN ANALYZE
- Developing the application logic for article creation and search
- Preparing the final report and supporting documentation

Initial References:

1. PostgreSQL Global Development Group. PostgreSQL Documentation: Full Text Search.
<https://www.postgresql.org/docs/current/textsearch.html>
2. PostgreSQL Global Development Group. PostgreSQL Documentation: GIN Indexes.
<https://www.postgresql.org/docs/current/gin.html>
3. Momjian, B. The Internals of PostgreSQL.
<https://www.interdb.jp/pg/>